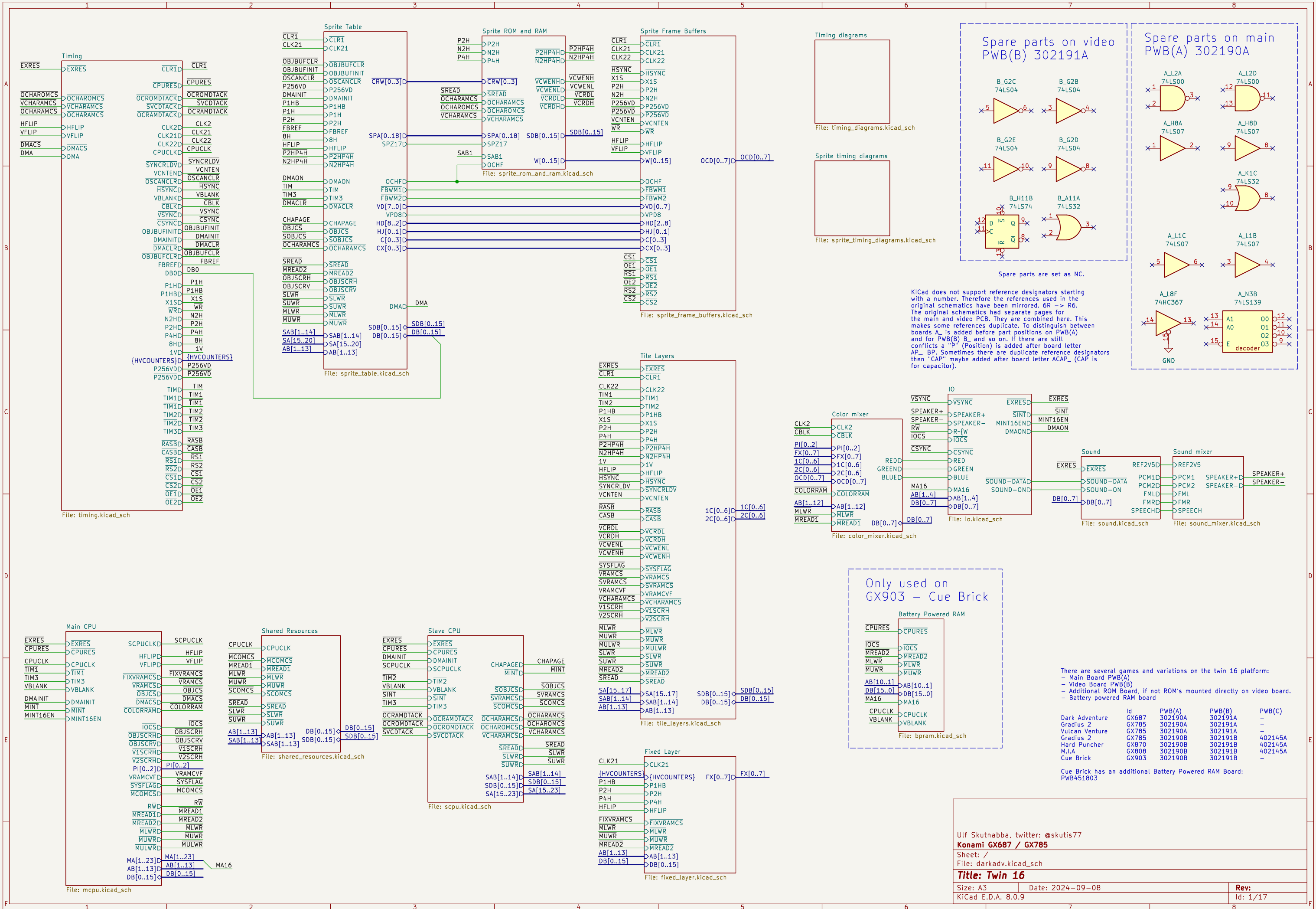
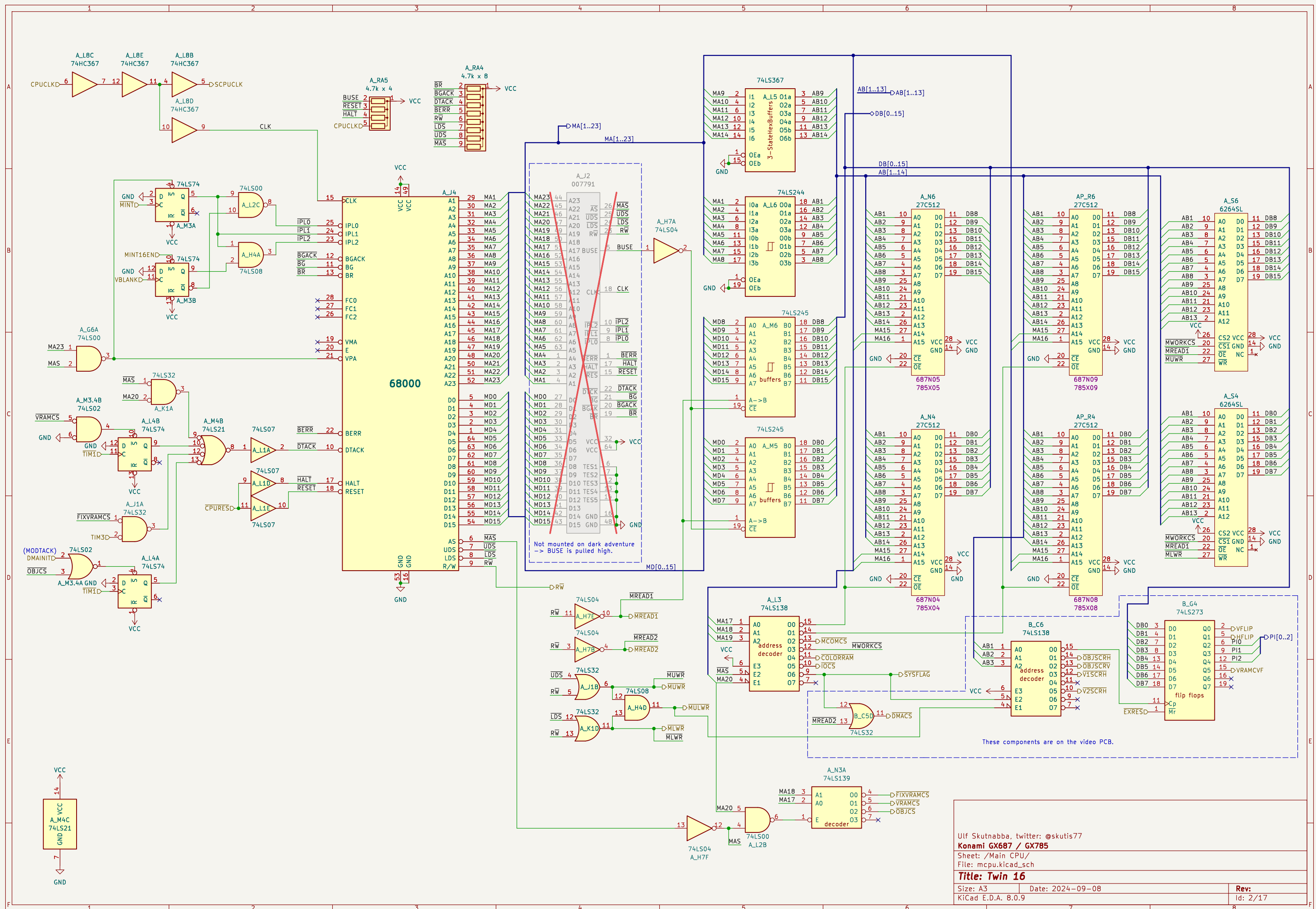
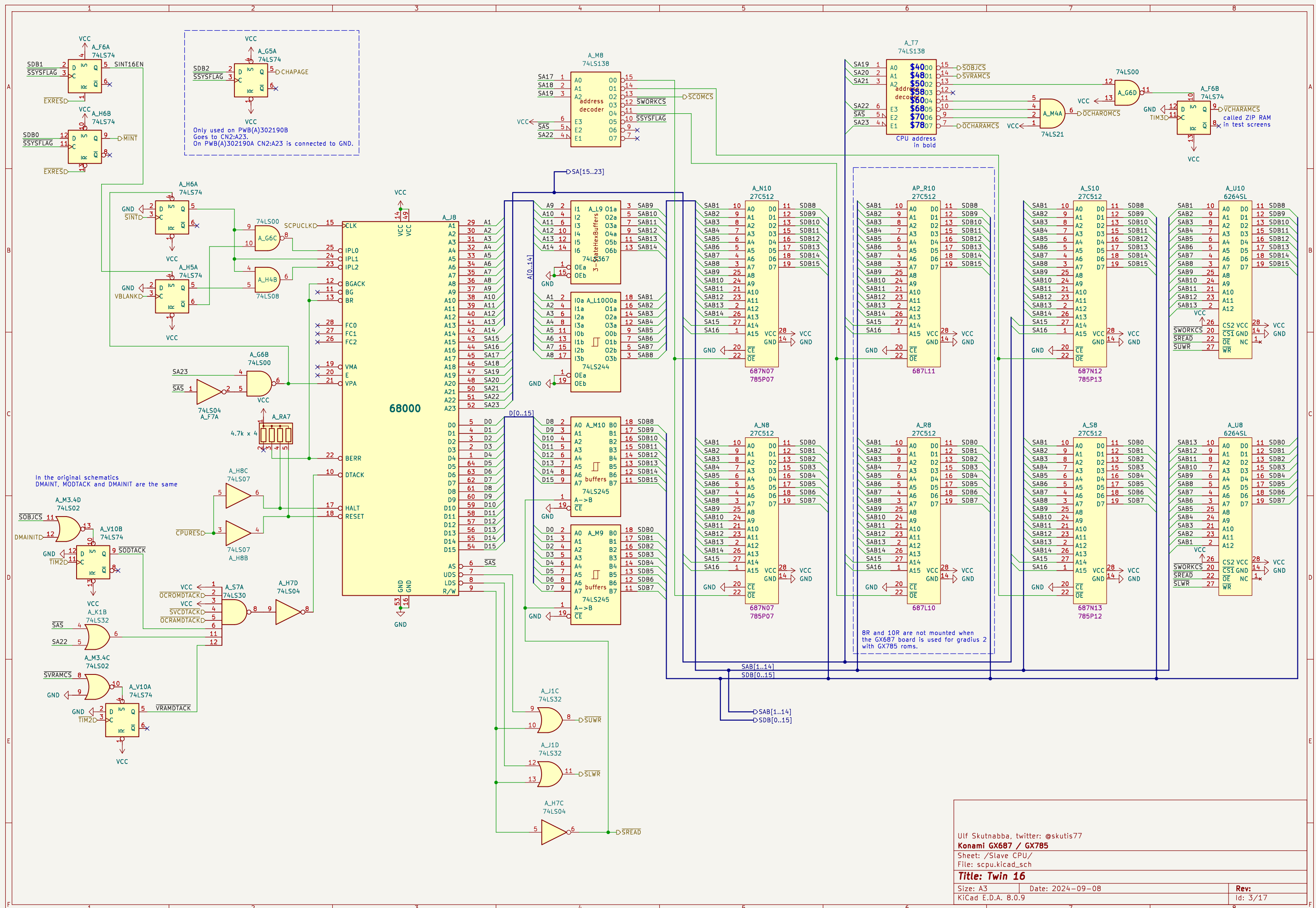




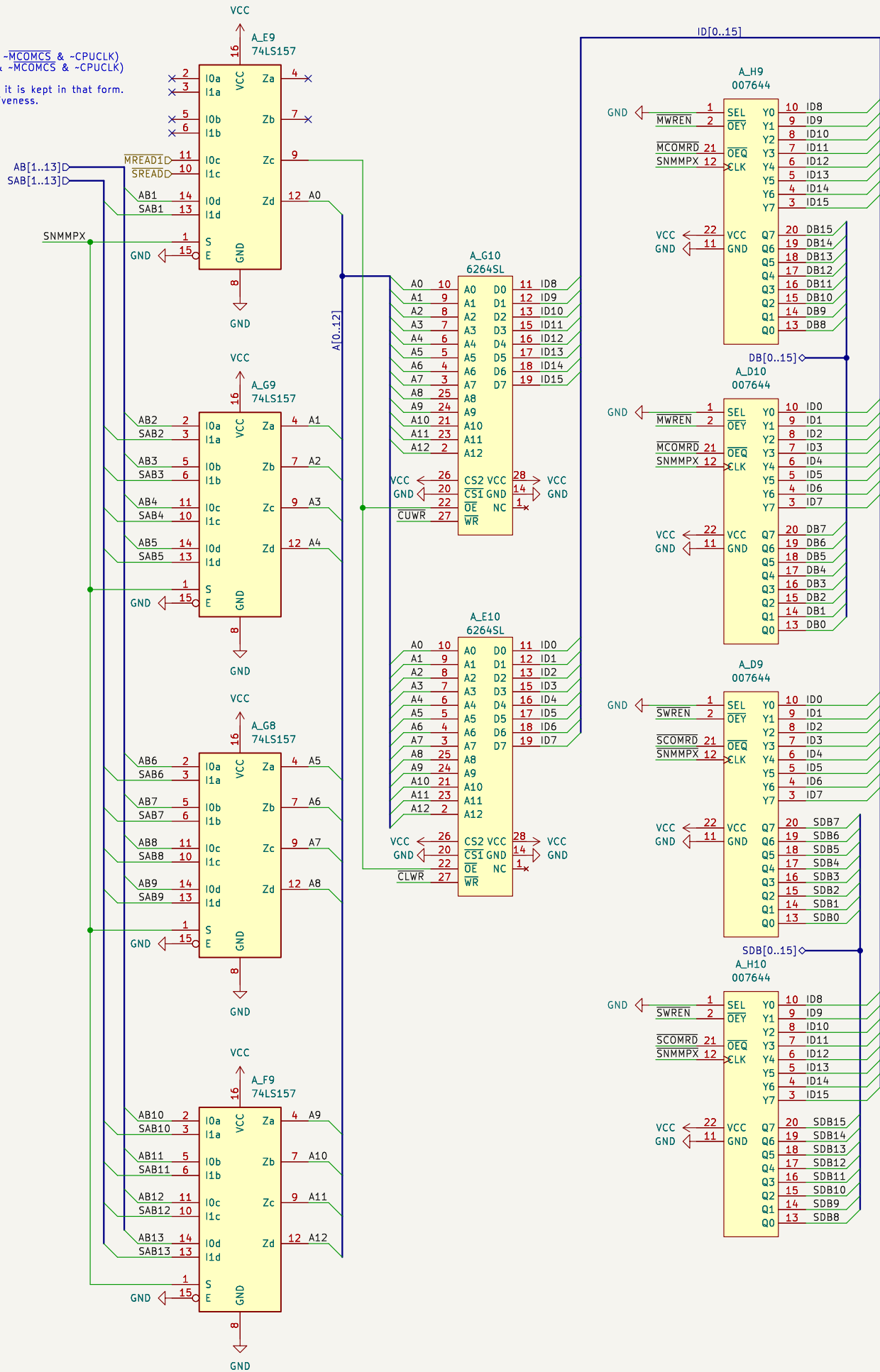
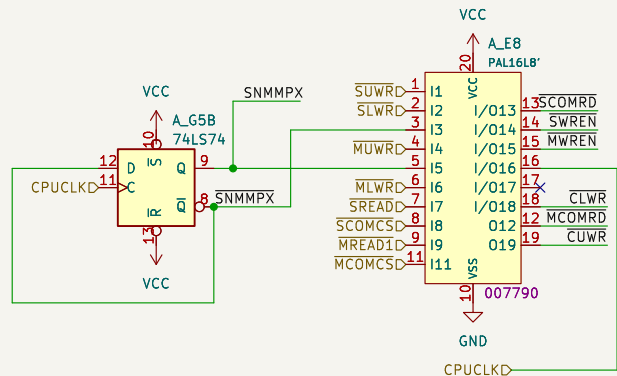




Konami 007790

```
MCOMRD = ~(-MREAD1 & ~MCOMCS) = MREAD1 | MCOMCS
SCOMRD = ~(-SREAD & ~SCOMCS)
SWREN = ~(-SNMMPX & SREAD & ~SCOMCS)
MWREN = ~(-SNMMPX & MREAD1 & ~MCOMCS)
CLWR = ~(-SLWR & ~SNMMPX & SNMMPX & SREAD & ~SCOMCS & ~CPUCLK | ~SNMMPX & ~MLWR & MREAD1 & ~MCOMCS & ~CPUCLK)
CUWR = ~(-SUWR & ~SNMMPX & SNMMPX & SREAD & ~SCOMCS & ~CPUCLK | ~MUWR & ~SNMMPX & MREAD1 & ~MCOMCS & ~CPUCLK)
```

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form. The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.



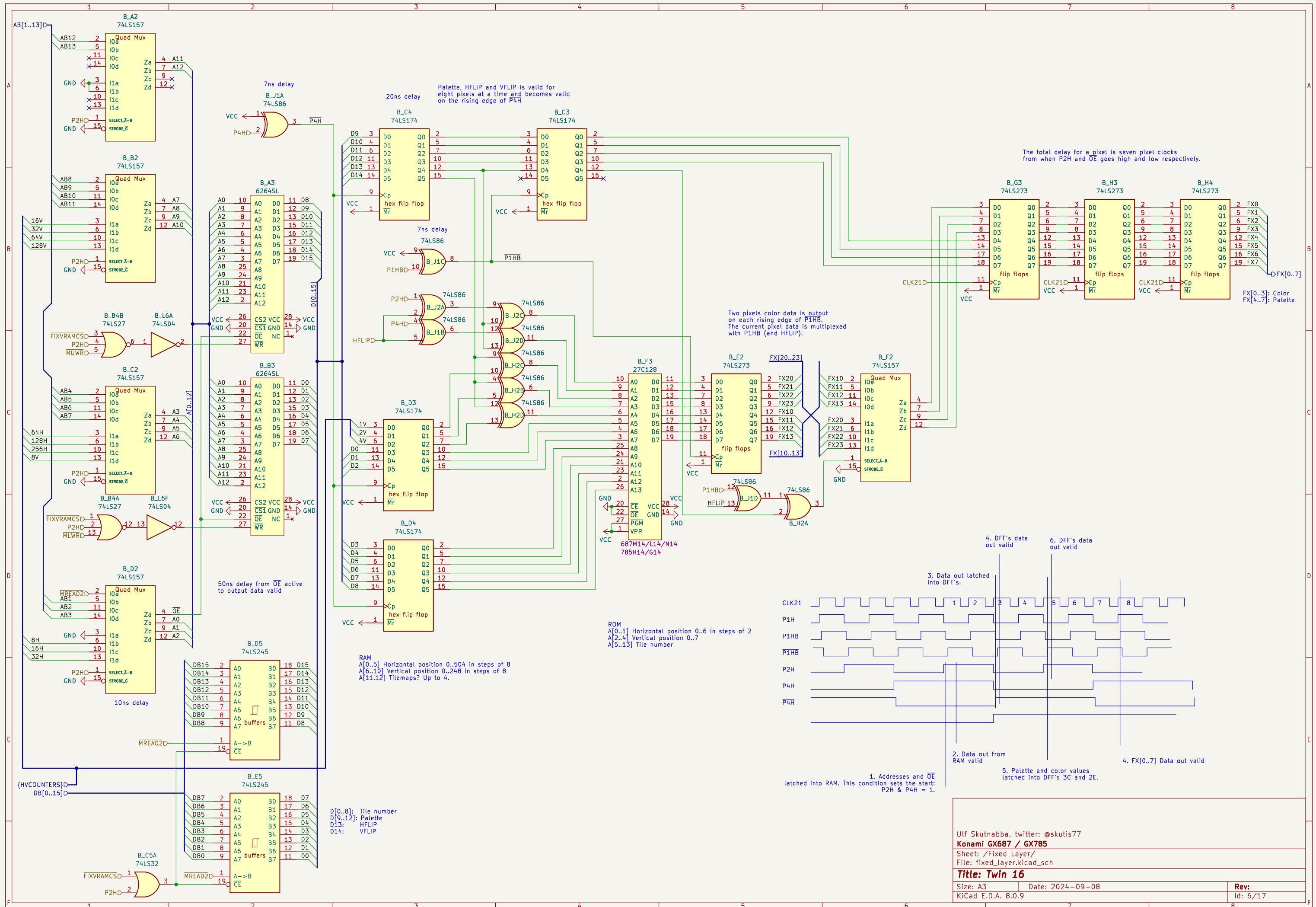
This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.





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Konami GX687 / GX785

Sheet: /Fixed Layer/

File: fixed_layer.kicad_sch

Title: Twin 16

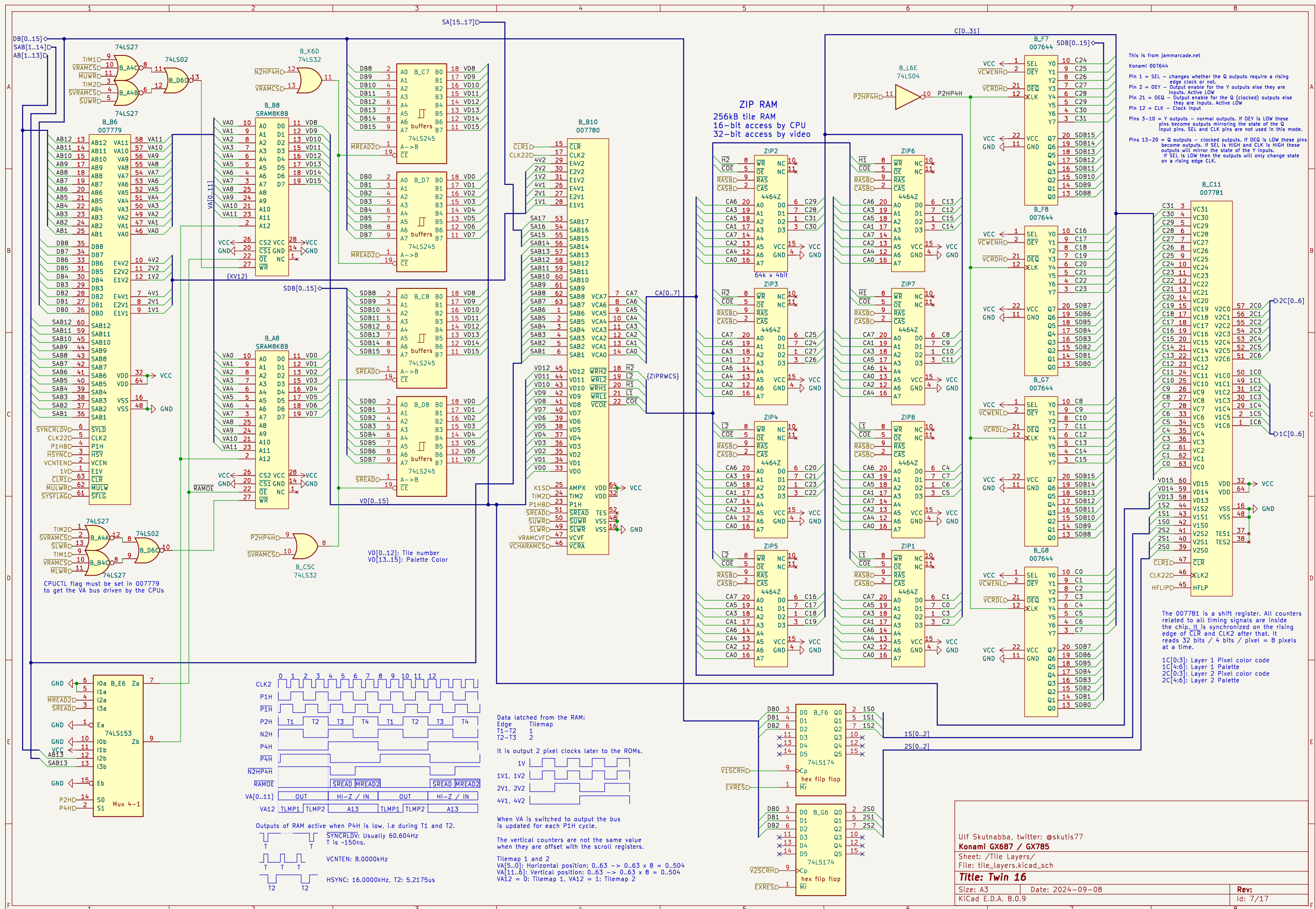
Size: A3

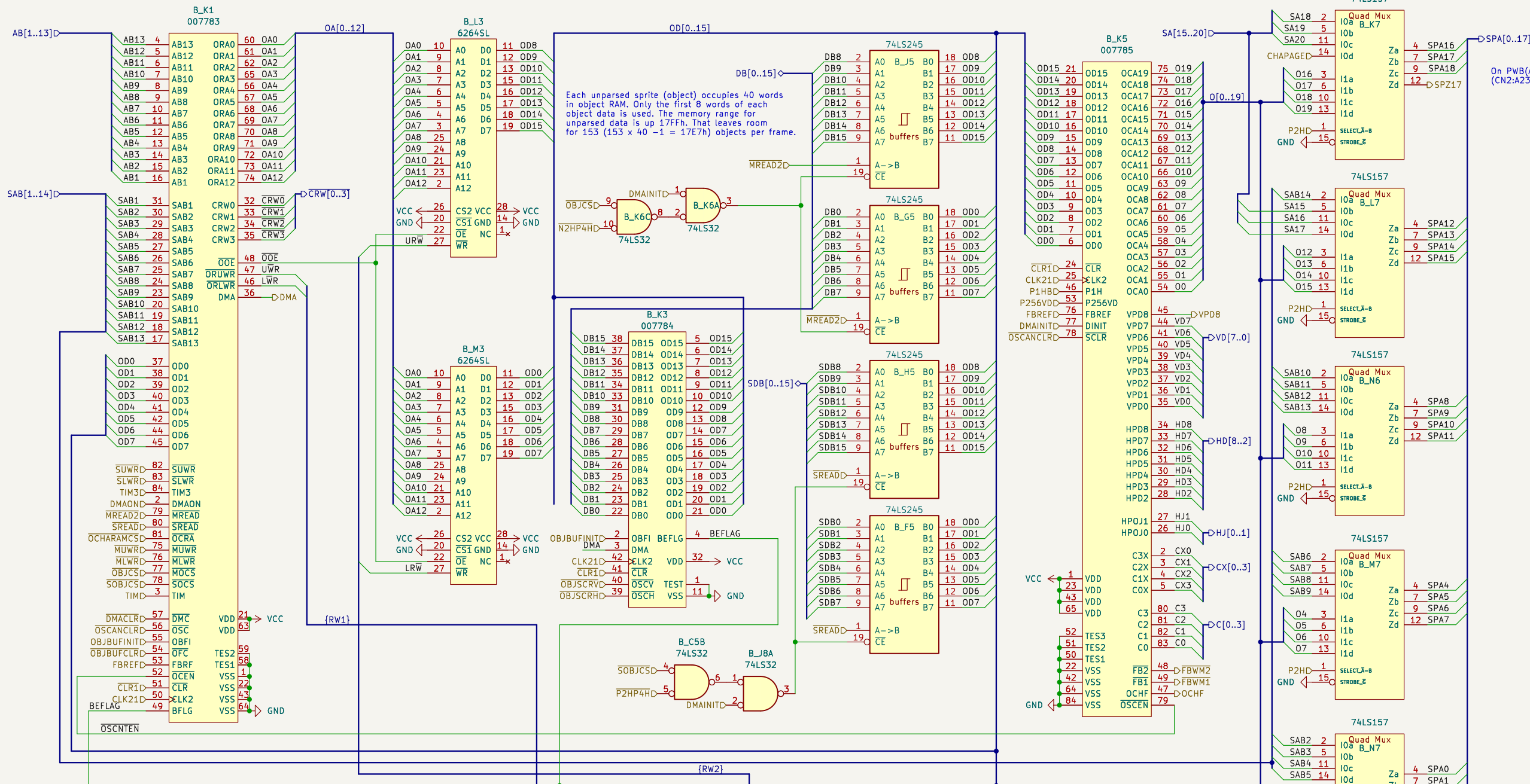
Date: 2024-09-08

Rev:

KiCad E.D.A. 8.0.9

Id: 6/17





007784 Sprite parser

- subtracts the scroll value for the sprites from current coordinates for the sprites.
- When DMA goes high, the 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM. It will compress the eight words into four words and also subtract the x- and y-scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
- Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..

Scroll Register

OBJSCRV	Sprite horizontal scroll
OBJSCRH	Sprite vertical scroll

Data is latched in the scroll registers when the signals OBJSCRV and OBJSCRH go low.

007793

CLK21' = CLK21

LRW = -(DMAONS & OBJBUFINIT & CLK21' | BEFLAG & DMA & P1H & DMAONS & CLK21' & (-BH & HFLIP | 8H & -HFLIP) -LRW & -DMAINIT)

URW = -(DMAONS & OBJBUFINIT & CLK21' | BEFLAG & DMA & P1H & DMAONS & CLK21' & (BH & -HFLIP | -BH & HFLIP) -URW & -DMAINIT)

The sprite memory is controlled in three separate cases:

1. OBJBUFINIT
2. BEFLAG & DMA
3. -DMAINIT

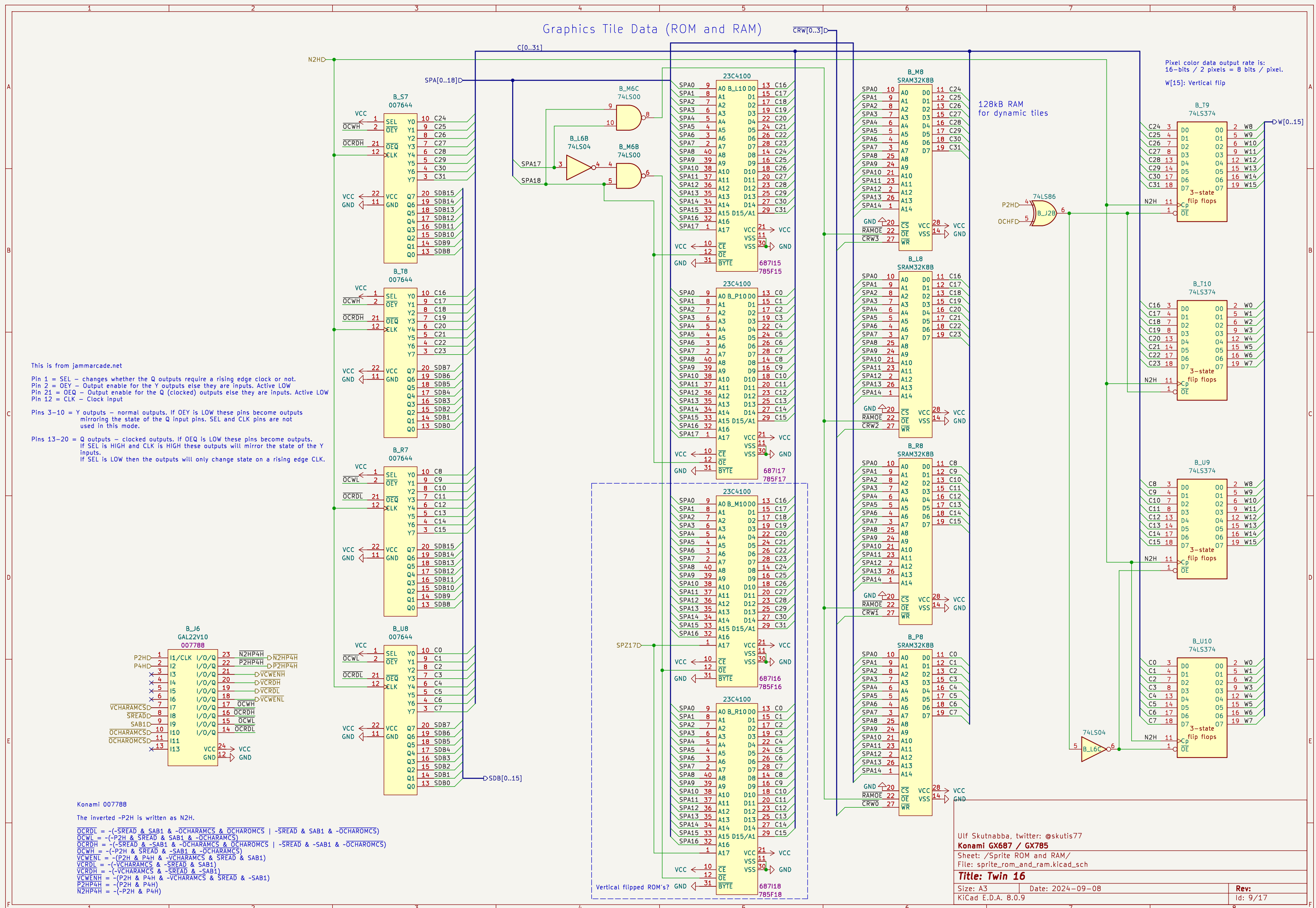
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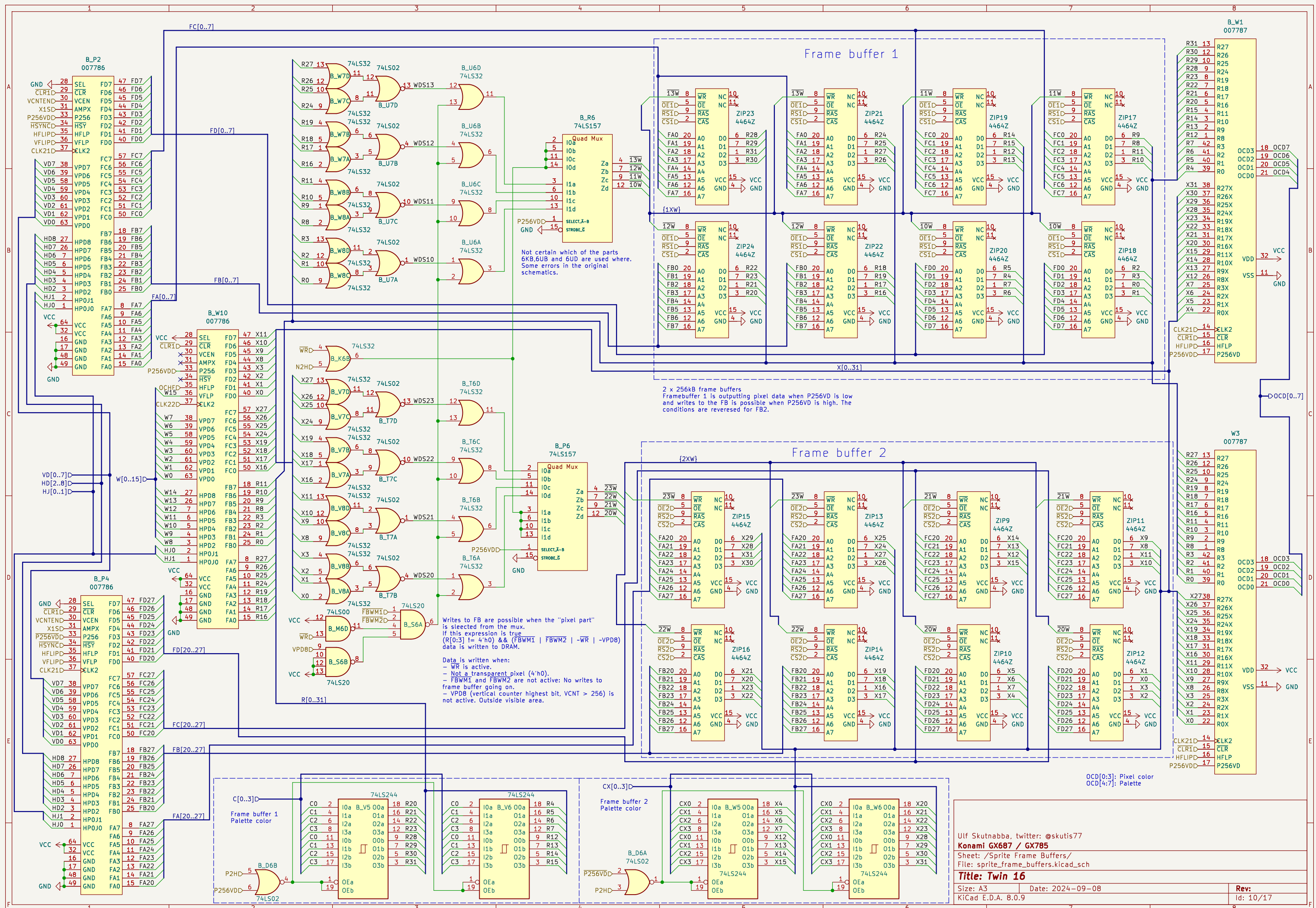
Konami GX687 / GX785

Sheet: /Sprite Table/
File: sprite_table.kicad_sch

Title: Twin 16

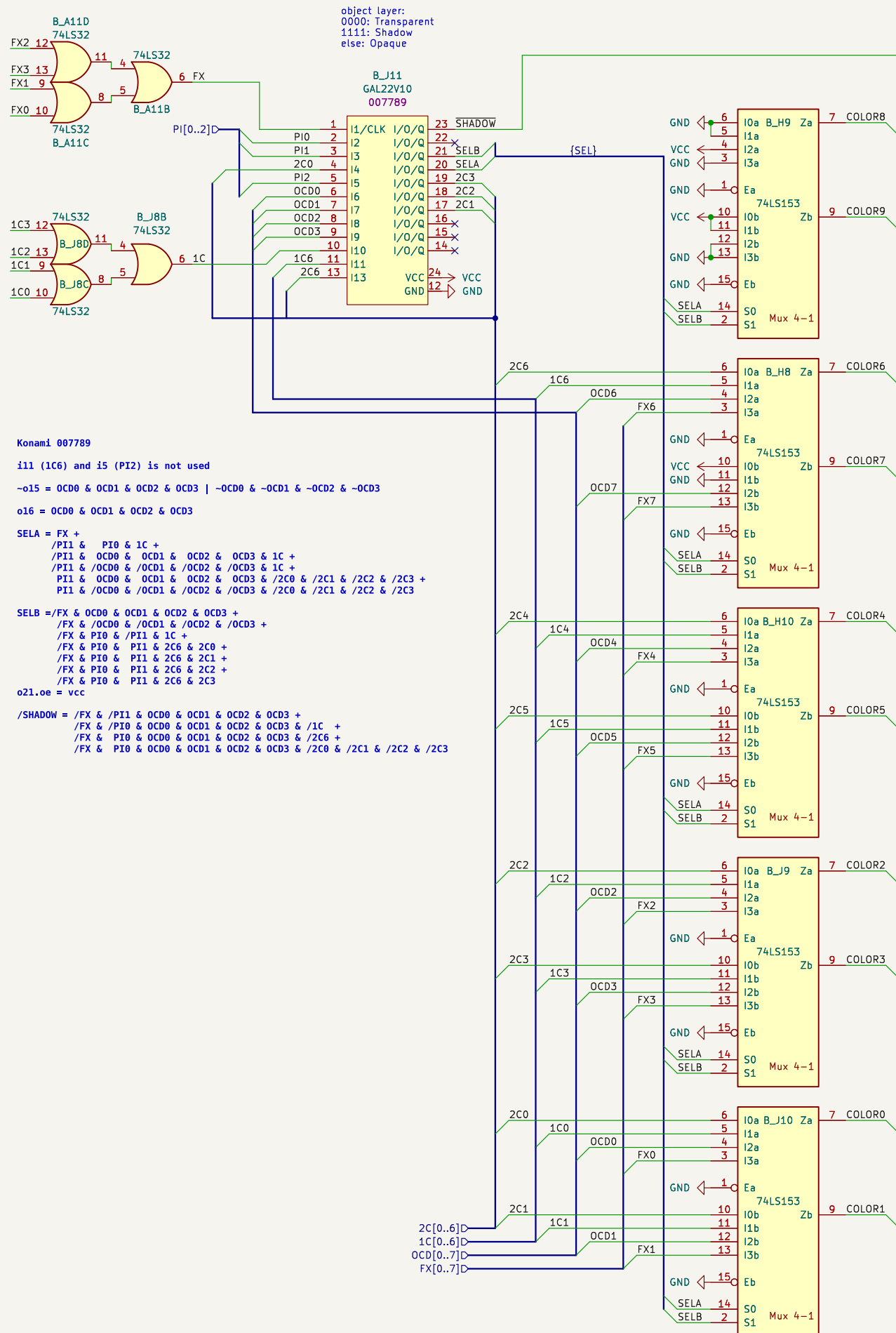
Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 8/17





Components are on Video PCB

Components are on Main PCB



Konami 007789

i11 (1C6) and i5 (PI2) is not used

~o15 = OCD0 & OCD1 & OCD2 & OCD3 | ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3

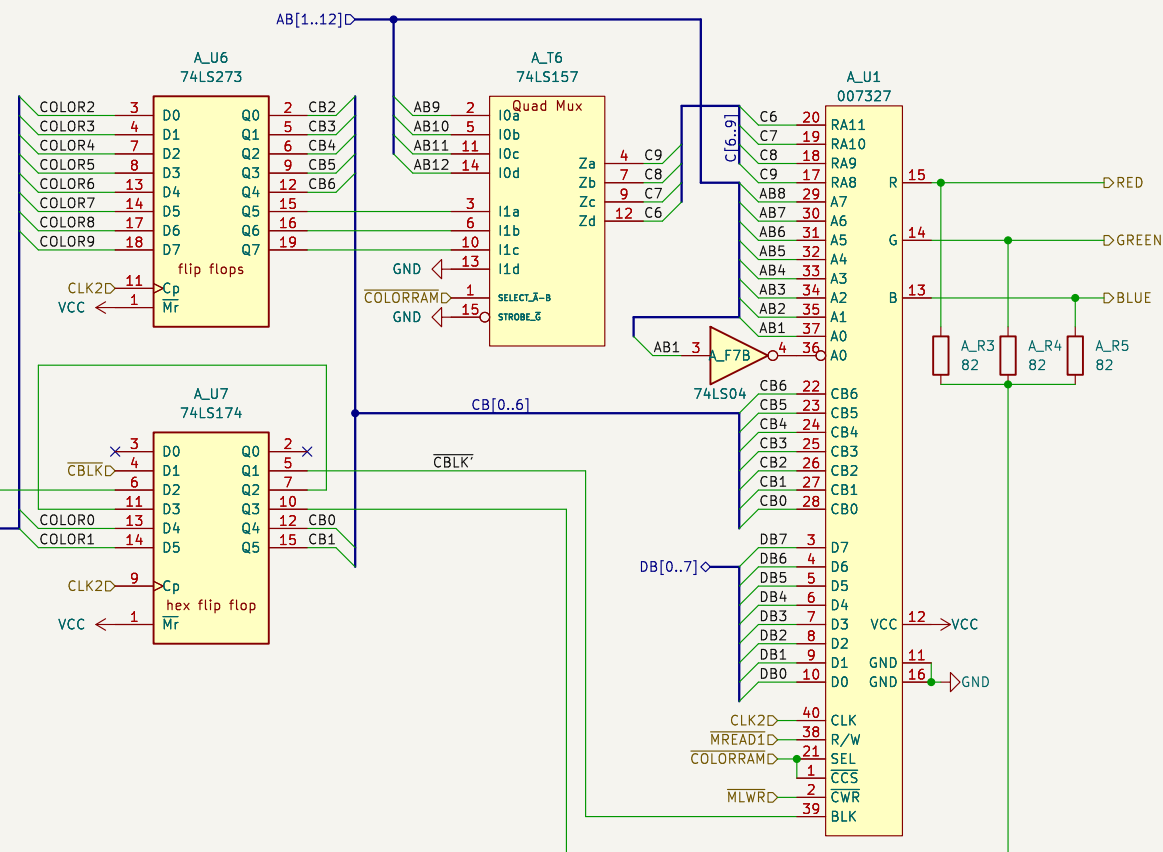
o16 = OCD0 & OCD1 & OCD2 & OCD3

SELA = FX +
/PI1 & PI0 & 1C +
/PI1 & OCD0 & OCD1 & OCD2 & OCD3 & 1C +
/PI1 & /OCD0 & /OCD1 & /OCD2 & /OCD3 & 1C +
PI1 & OCD0 & OCD1 & OCD2 & OCD3 & /2C0 & /2C1 & /2C2 & /2C3 +
PI1 & /OCD0 & /OCD1 & /OCD2 & /OCD3 & /2C0 & /2C1 & /2C2 & /2C3

SELB = /FX & OCD0 & OCD1 & OCD2 & OCD3 +
/FX & /OCD0 & /OCD1 & /OCD2 & /OCD3 +
/FX & PI0 & /PI1 & 1C +
/FX & PI0 & PI1 & 2C6 & 2C0 +
/FX & PI0 & PI1 & 2C6 & 2C1 +
/FX & PI0 & PI1 & 2C6 & 2C2 +
/FX & PI0 & PI1 & 2C6 & 2C3

o21.o0 = vcc

/SHADOW = /FX & /PI1 & OCD0 & OCD1 & OCD2 & OCD3 +
/FX & /PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /1C +
/FX & PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /2C6 +
/FX & PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /2C0 & /2C1 & /2C2 & /2C3



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Konami GX687 / GX785

Sheet: /Color mixer/

File: color_mixer.kicad_sch

Title: Twin 16

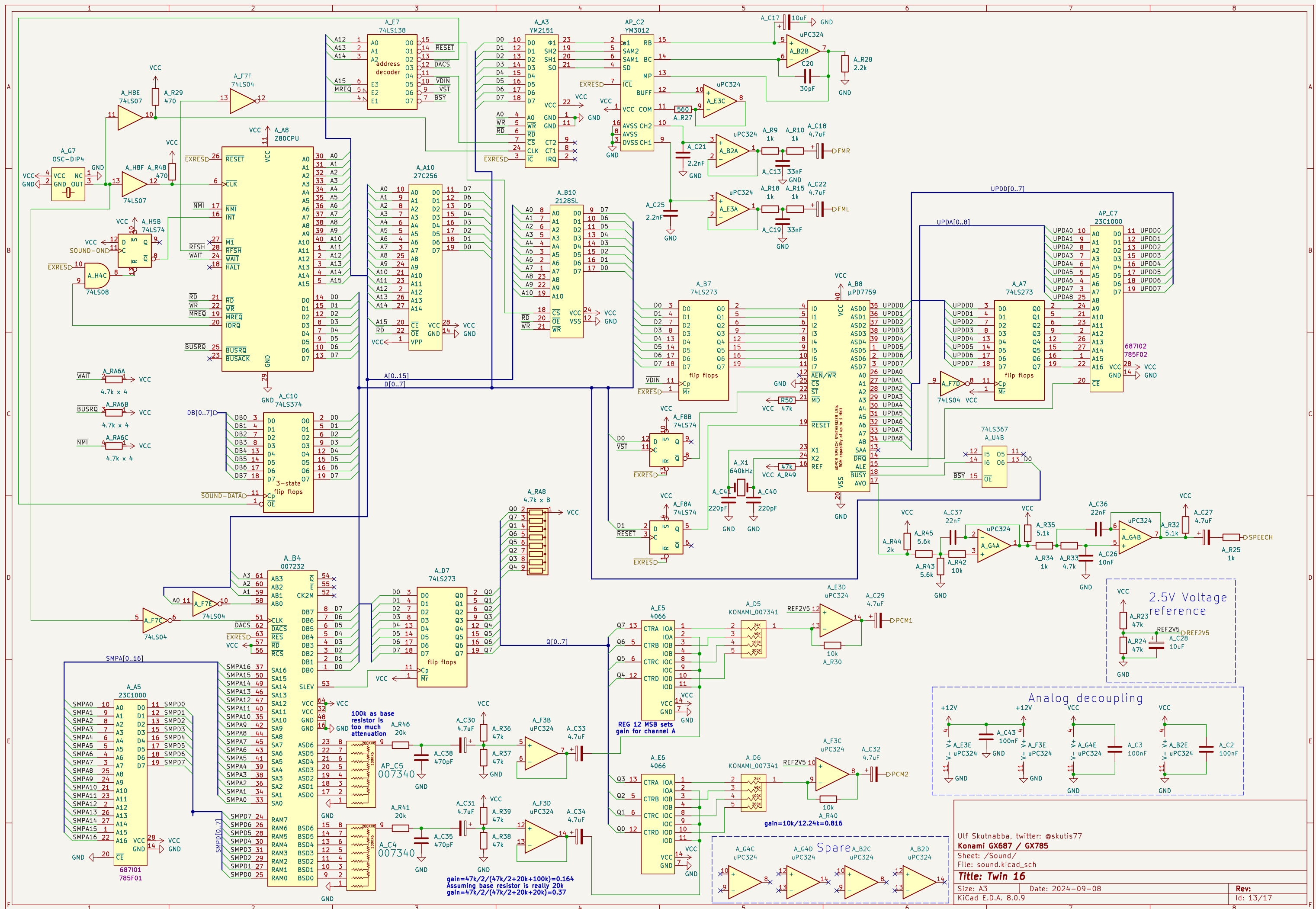
Size: A3

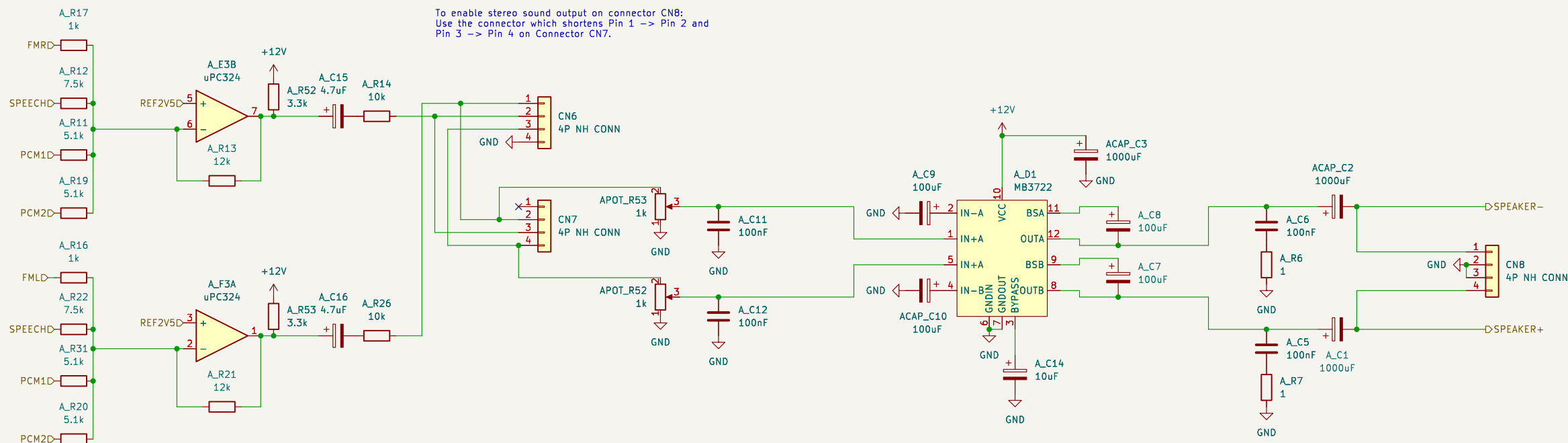
Date: 2024-09-08

Rev:

KiCad E.D.A. 8.0.9

Id: 11/17





To enable stereo sound output on connector CN8:
Use the connector which shortens Pin 1 -> Pin 2 and
Pin 3 -> Pin 4 on Connector CN7.

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Konami GX687 / GX785

Sheet: /Sound mixer/

File: sound_mixer.kicad_sch

Title: Twin 16

Size: A3

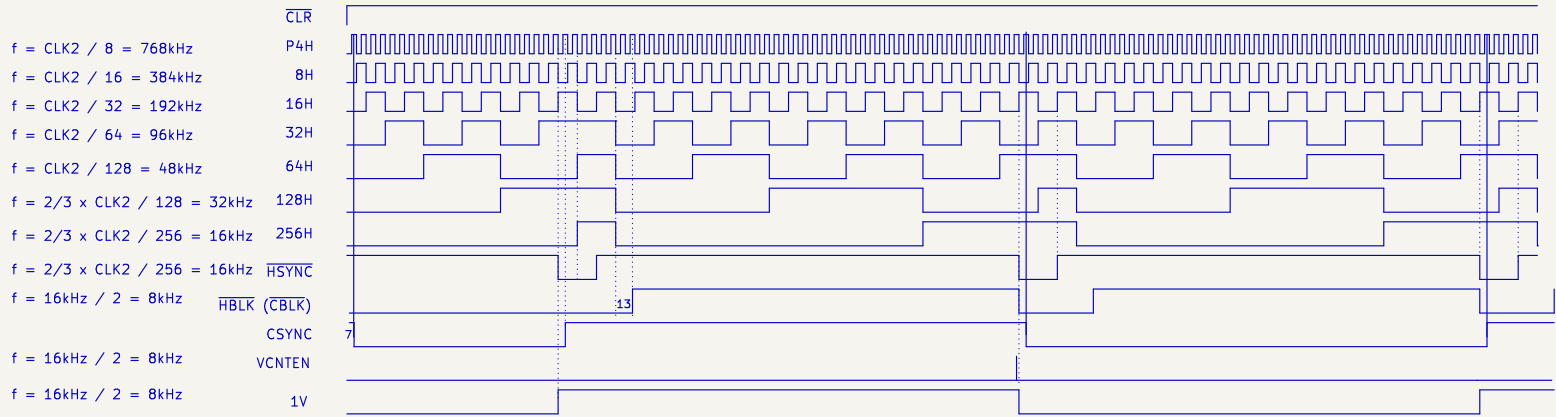
Date: 2024-09-08

Rev:

KiCad E.D.A. 8.0.9

Id: 14/17

Horizontal signals

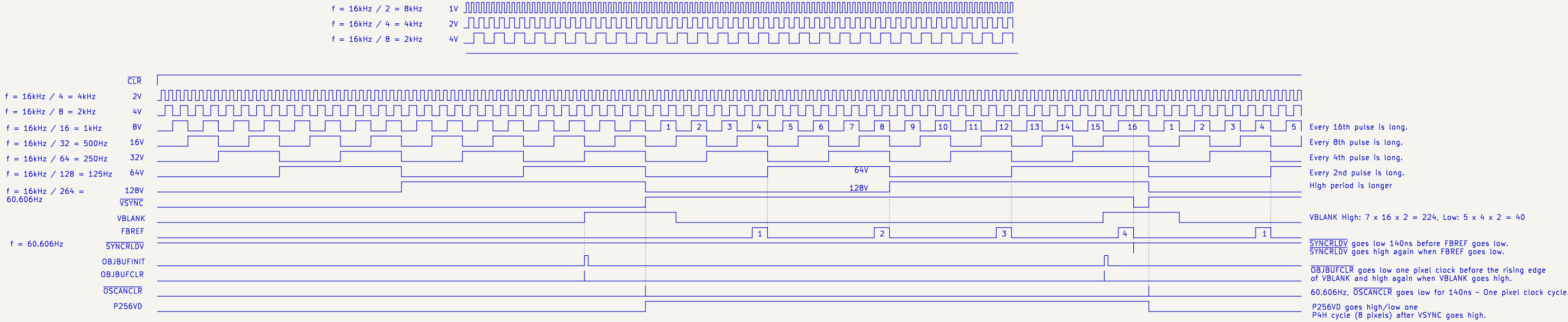


- The first VCNTEN is skipped after reset. It goes low 140ns before HSYNC goes low, and high again when HSYNC goes low. VCNTEN is active right before every second falling edge of HSYNC.

- CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The first HSYNC is at HCOUNT 175 after reset.

Vertical signals



Every 16th pulse is long.
Every 8th pulse is long.
Every 4th pulse is long.
Every 2nd pulse is long.
High period is longer

VBLANK High: $7 \times 16 \times 2 = 224$, Low: $5 \times 4 \times 2 = 40$

SYNCRLDV goes low 140ns before FBREF goes low. SYNCRLDV goes high again when FBREF goes low.

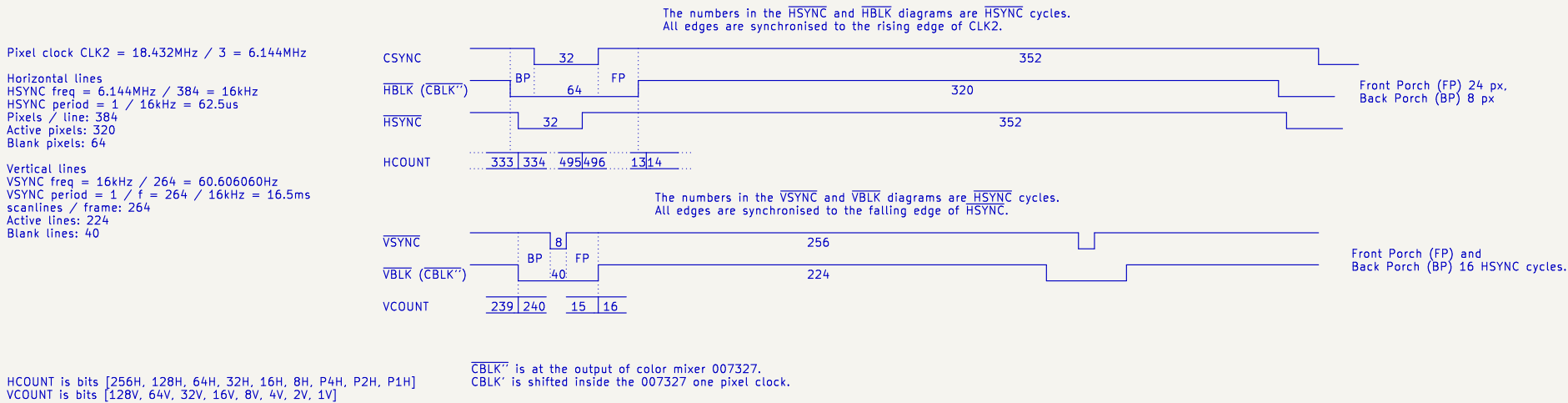
OBJBUFCLR goes low one pixel clock before the rising edge of VBLANK and high again when VBLANK goes high.

60.606Hz, OSCANCLR goes low for 140ns - One pixel clock cycle.

P256VD goes high/low one P4H cycle (8 pixels) after VSYNC goes high.

FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

Horizontal and vertical synch timing diagrams



Pixel clock CLK2 = $18.432\text{MHz} / 3 = 6.144\text{MHz}$

Horizontal lines
HSYNC freq = $6.144\text{MHz} / 384 = 16\text{kHz}$
HSYNC period = $1 / 16\text{kHz} = 62.5\mu\text{s}$
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

Vertical lines
VSYNC freq = $16\text{kHz} / 264 = 60.606060\text{Hz}$
VSYNC period = $1 / f = 264 / 16\text{kHz} = 16.5\text{ms}$
scanlines / frame: 264
Active lines: 224
Blank lines: 40

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

CBLK" is at the output of color mixer 007327.
CBLK' is shifted inside the 007327 one pixel clock.

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Konami GX687 / GX785
Sheet: /Timing diagrams/
File: timing_diagrams.kicad_sch

Title: Twin 16

Size: A3
KiCad E.D.A. 8.0.9

Date: 2024-09-08

Rev:
Id: 15/17

